

# A Modernization of the DreamWorks Feather System

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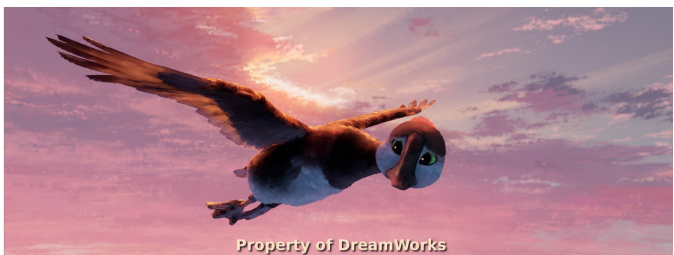


Figure 1: Feathers from the films *Kung Fu Panda 4* and *The Wild Robot* created using the DreamWorks Feather System.

## ABSTRACT

The DreamWorks Feather System is a toolset used to design feathered characters. It enables artists to manage all aspects of feathers from card layout through shot work by using intuitive tools and automated pipeline processes. While the previous system proved inaccessible for some due to its highly technical nature, these new developments have enabled dozens of artists to successfully groom characters and run their feathers in shots. An initial feather layout can now be created in a matter of minutes, with a suite of comprehensive tools allowing for further refinement and precise control over feather card layout, feather curves, and animated motion.

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## 1 MOTIVATION

Grooming and moving feathers at DreamWorks has a long history of innovation but has been a technically complicated and specialized process, involving the expertise of multiple departments. Recent

productions *Kung Fu Panda 4* and *The Wild Robot* brought feathered characters back to the forefront, making us hatch up new plans to re-imagine the existing toolset to be approachable to new artists.

## 2 FEATHER CARD GROOMING

At DreamWorks, ‘body feathers’ are procedurally scattered through our card grooming tool while ‘acting feathers’, such as wing feathers, are modeled and rigged through the standard character pipeline. Building off *Hummingbird* [Augello et al. 2019], the new feather card grooming tool has improved the system in significant ways.

### 2.1 Feather Direction Controls

A new tool was developed for controlling the orientation of feathers, aiming to make this experience more intuitive while maintaining topology independence. Feathers can now be oriented along linear or radial target directions, or be advected along custom drawn curves. Paint masks and bounding objects allow artists to define which regions of the groom to modify, while various mirroring options enable these adjustments to be quickly applied to other areas. Ease of use was further enhanced through the addition of customizable visualizers, equipping artists to clearly see their changes.

### 2.2 Feather Heroing

Art direction on *The Wild Robot* necessitated precise feather placement and thus an environment for unrestricted artistic control was needed. Instead of using only paint maps and controls for procedural scattering, artists are able to freely add, remove, or reposition feathers to accomplish their goals. Additional heroing options include drawing tools for placing feathers on the skin and generating feathers in a cylindrical pattern around the skin.

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Figure 2: Laminated feather cards visualized in a shot

### 2.3 Lamination Function Generation

A primary issue of *Hummingbird* was the technically complex nature of the lamination function [Weber and Gornowicz 2009], which uses a custom language to control the layering of feathers. In order to address this issue, the new generation of feathers provides artist-friendly controls for generating this lamination function automatically. Building off of *Hummingbird*'s paintable loft attributes, artists can further adjust these parameters with animation values from the rig, multipliers, and random loft offsets. These settings can then be read into the lamination function in real-time, allowing for quick visual feedback and iteration. Thus, instead of having to write several hundreds of lines by hand, a custom lamination script with accurate syntax is now generated nearly instantaneously.

### 3 FEATHER CURVES PROCEDURAL

The DreamWorks Hair System, *Willow*, was updated to support the creation of feathers, taking advantage of all of the previous features and our existing pipeline. Feathers in *Willow* use a closest-point wrap method to bind the reference hair data to the animated card geometry for each feather. Feathers are procedurally generated in *Willow* keeping the disk memory low. The feather quill and barb design process contains a wide range of artistic control allowing artists to create unique feathers on a character and transfer the settings to other feather assets. Look Development artists can then use existing hair shading techniques. The feathers are not instanced, but each feather follows a weighted interpolation of the defined feather types, which are not limited by quantity like prior implementations. This makes it easy to define a smooth transition between feathers using painted maps on the skin or on the feather cards themselves.

### 4 VISUALIZATION

A custom GPU visualizer was built off the WIZ system [Somasundaram and Shay 2022] to display animated high resolution feathers' curves in real time in the 3-D viewport. New viewport visualizers can be used to display feather attributes as maps as well as assign textures to feathers for accurate visual feedback. A subset of feathers can be isolated for precise control and faster iteration.

### 5 FEATHER SHOT WORK

The new feather shot tool takes any feather groom and presents an environment for artists to manipulate the feathers within their shots. In addition to *Hummingbird*'s simulation system, new finaling tools enable artists to procedurally reshape, reposition, animate feathers, add wind, remove jitter, and deform feathers using the

existing curve deformation tools. These tools are used in the same environment as the acting feathers and skin finaling, consolidating the artist experience into a single scene.

## 6 CROWDS

To save on render resources, Crowds pre-generates the feathers and bakes them into Alembic geometric curves. They then cull down the density of the barbs, increase the radius, and randomly delete a fraction of points along each curve. For even more savings, textured cards can also be used for renders. This geometry is then switched out in USD and the material shader network is unaffected.

## 7 PIPELINE

A number of pipeline enhancements brought the new tools together. A new publishing system saves versioned feather grooms. A new setup tool facilitates easily configuring a scene file for either card grooming or look development. A management system ensures the grooms can get all the latest tool updates without data loss. Look Development and CFX have different internal workflows, and new pipeline tools allow for a parallel workflow between the two departments with minimal differences in the artist experience.

Running feather grooms in shots also required pipeline enhancements. We attached an automated groom execution step to Animation's model caching, which allows Animation to see feathers in their renderers, and serves as a first pass for Lighting. If the feathers are problematic, or the shot has more complex needs, CFX can use their standard shot workflow used for clothing and hair to adjust the feathers and apply specialized effects such as wind.

## 8 CONCLUSION

Building up this new generation of feathers has been a massive cross-departmental effort. The comprehensive card layout toolset, new curves procedural, real-time viewport visualization, shot-specific feather controls, crowd LOD features, and automated pipeline have significantly improved the workflow and enabled dozens of additional artists to use feathers. This new system was successfully used on hundreds of shots on *Kung Fu Panda 4* and *The Wild Robot*, with improvements being implemented as new productions take flight.

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